

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester I Examination, 2016-17

Paper No. MTB-101: Calculus I

Time: 3 Hours

Full Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Show that $\lim_{n \rightarrow \infty} \frac{1+3+5+\dots+(2n-1)}{n^2} = 1$.
- (b) Show that the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ is not convergent.
- (c) Test the continuity of the function defined by

$$f(x) = \begin{cases} \frac{|x-a|}{x-a}, & x \neq a \\ 1, & x = a \end{cases}$$

at $x = a$. If discontinuous, write the type of discontinuity at $x = a$.

- (d) Find the value of the derivative of the function $f(x) = |x-1| + |x-3|$ at $x = 2$.
- (e) If $y = \cos^2 x$, then find y_n .
- (f) If $e^x = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$, then find the value of a_3 .
- (g) By using the definition of limit, show that $\lim_{x \rightarrow 2} \frac{1}{(x-2)^2} \rightarrow \infty$.
- (h) Discuss the applicability of Rolle's theorem to the function:

$$f(x) = \begin{cases} x^2 + 1, & 0 \leq x < 1 \\ 3 - x, & 1 \leq x \leq 2 \end{cases}$$

- (i) Find all the asymptotes of the curve $y^2(x^2 - y^2) = a^2(x^2 + y^2)$, $a > 0$.
- (j) Find the nature of the multiple point $(0, 0)$ on the curve $x^2(x - y) + y^3 = 0$.
- (k) By using definite integration, find the limit:

$$\lim_{n \rightarrow \infty} \left[\frac{\sqrt{n^2 - 1^2}}{n^2} + \frac{\sqrt{n^2 - 2^2}}{n^2} + \dots + \frac{\sqrt{n^2 - (n-1)^2}}{n^2} \right]$$

Question 2.

- (a) Define the convergence of a sequence (x_n) . By using the definition, show that the sequence (x_n) defined by $x_n = \sqrt{n+1} - \sqrt{n}$ is convergent. [4]

(b) Test the convergence of the series:

$$\frac{1}{2} + \frac{1 \cdot 2}{3 \cdot 5} + \frac{1 \cdot 2 \cdot 3}{3 \cdot 5 \cdot 7} + \frac{1 \cdot 2 \cdot 3 \cdot 4}{3 \cdot 5 \cdot 7 \cdot 9} + \dots$$

[4]

(c) Test the convergence of the series:

$$\frac{1}{2} + \frac{2}{5} + \frac{3}{10} + \dots + \frac{n}{n^2 + 1} + \dots$$

[4]

Question 3.

(a) Prove that every monotonic increasing sequence which is bounded above is convergent. [6]

(b) If $f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$, then show that the function is differentiable at $x = 0$ but the derivative is not continuous at $x = 0$. [6]

Question 4.

(a) Let $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$. Show that $\lim_{x \rightarrow a} f(x)g(x) = lm$. [6]

(b) State and prove the intermediate value theorem for continuous functions. [6]

Question 5.

(a) State and prove Lagrange's mean value theorem. Provide its geometrical interpretation. [6]

(b) If $y^{1/m} + y^{-1/m} = 2x$ for $m \in \mathbb{N}$, then prove that:

$$(x^2 - 1)y_{n+2} + (2n + 1)xy_{n+1} + (n^2 - m^2)y_n = 0$$

[6]

Question 6.

(a) Find all the asymptotes of the curve:

$$x^3 - x^2y - xy^2 + y^3 + 2x^2 - 4y^2 + 2xy + x + y + 1 = 0$$

[6]

- (b) State and prove Taylor's theorem in finite form with Lagrange's form of remainder. [6]
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Question 7.

- (a) Trace the curve:

$$y^2(a+x) = x^2(3a-x)$$

[6]

- (b) By applying the definition of definite integration as the limit of a sum, evaluate:

$$\lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \cdots \left(1 + \frac{n}{n}\right) \right]^{1/n}$$

[6]